

Network Security & Cryptography Assignment 3 Solutions

Question 1: Purpose of S-boxes in DES [10 Marks]

The Substitution boxes (S-boxes) in DES serve several critical purposes:

Primary Functions:

- **Non-linear transformation:** S-boxes provide the only non-linear component in DES, making the cipher resistant to linear cryptanalysis
- **Confusion:** They obscure the relationship between plaintext, key, and ciphertext by substituting input bits with different output bits
- **Avalanche effect:** Small changes in input produce significant changes in output
- **Security strength:** S-boxes are the main source of DES's cryptographic strength

Technical Details:

- DES uses 8 S-boxes (S1 through S8)
- Each S-box takes 6 bits as input and produces 4 bits as output
- The substitution is based on predetermined lookup tables
- S-boxes compress 48 bits to 32 bits during the f-function

Security Significance: The S-boxes were designed with specific criteria to resist known attacks and provide maximum confusion, making DES secure against differential and linear cryptanalysis techniques available at the time of its design.

Question 2: Comparison of Private Key and Public Key Cryptography [10 Marks]

Aspect	Private Key Cryptography	Public Key Cryptography
Key Structure	Single shared secret key	Key pair (public + private keys)
Key Distribution	Secure key exchange required before communication	Public key can be distributed openly
Speed	Very fast encryption/decryption	Slower due to complex mathematical operations
Scalability	Poor scalability ($n(n-1)/2$ keys for n users)	Excellent scalability (2 keys per user)
Key Management	Difficult key management for large networks	Easier key management
Authentication	Cannot provide non-repudiation	Provides digital signatures and non-repudiation
Computational Complexity	Low computational overhead	High computational overhead
Key Length	Shorter keys (128-256 bits typically)	Longer keys (1024-4096 bits typically)
Primary Use	Bulk data encryption	Key exchange, digital signatures, small data
Examples	AES, DES, 3DES, Blowfish	RSA, ECC, Diffie-Hellman, DSA

Question 3: Achieving Confidentiality with Public Key Cryptography [10 Marks]

Confidentiality in public key cryptography is achieved through the following mechanism:

Basic Principle:

- Data encrypted with a public key can only be decrypted with the corresponding private key
- The private key is kept secret by the owner, ensuring only they can decrypt messages

Process for Confidentiality:

1. **Key Generation:** Receiver generates a public-private key pair
2. **Public Key Distribution:** Receiver shares their public key openly
3. **Encryption:** Sender encrypts the message using receiver's public key
4. **Transmission:** Encrypted message is sent over insecure channel
5. **Decryption:** Only the receiver can decrypt using their private key

Mathematical Foundation:

- Based on mathematical trapdoor functions (easy one way, hard reverse)

- RSA uses integer factorization problem
- ECC uses discrete logarithm problem

Practical Implementation:

- Often used in hybrid systems (encrypt symmetric key with public key)
- Ensures that even if communication is intercepted, confidentiality is maintained
- No need for prior secure key establishment between parties

Security Guarantee: Confidentiality is mathematically guaranteed as long as the private key remains secret and the underlying mathematical problem remains computationally infeasible.

Question 4: Achieving Authentication with Public Key Cryptography [10 Marks]

Authentication in public key cryptography is achieved through digital signatures:

Basic Principle:

- Data signed with a private key can be verified using the corresponding public key
- Only the private key holder can create valid signatures

Authentication Process:

1. **Message Preparation:** Sender creates message to be authenticated
2. **Hash Generation:** Message is hashed using cryptographic hash function
3. **Signature Creation:** Hash is encrypted with sender's private key
4. **Transmission:** Message and signature are sent together
5. **Verification:** Receiver decrypts signature with sender's public key
6. **Authentication:** If decrypted hash matches message hash, authentication succeeds

Types of Authentication Provided:

- **Entity Authentication:** Proves identity of the sender
- **Message Authentication:** Ensures message came from claimed sender
- **Non-repudiation:** Sender cannot deny sending the message
- **Integrity:** Ensures message hasn't been tampered with

Technical Implementation:

- Common algorithms: RSA signatures, DSA, ECDSA
- Often combined with hash functions (SHA-256, SHA-3)
- May use certificates for public key authentication

Security Features:

- Prevents impersonation attacks
 - Provides proof of origin
 - Enables secure communication without prior shared secrets
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Question 5: RSA Encryption/Decryption of 'HELLO' [20 Marks]

Given: $p = 12347$, $q = 181$, $e = 13$

Step 1: Calculate n and $\phi(n)$

- $n = p \times q = 12347 \times 181 = 2,234,807$
- $\phi(n) = (p-1)(q-1) = 12346 \times 180 = 2,222,280$

Step 2: Calculate private key d

- $d = e^{-1} \bmod \phi(n)$
- Using Extended Euclidean Algorithm: $d = 171,329$
- Verification: $(13 \times 171,329) \bmod 2,222,280 = 1 \checkmark$

Step 3: Convert 'HELLO' to numeric values

- $H = 72$, $E = 69$, $L = 76$, $L = 76$, $O = 79$

Step 4: Encryption ($C = M^e \bmod n$)

- $H: 72^{13} \bmod 2,234,807 = 1,317,460$
- $E: 69^{13} \bmod 2,234,807 = 1,261,239$
- $L: 76^{13} \bmod 2,234,807 = 1,914,022$
- $L: 76^{13} \bmod 2,234,807 = 1,914,022$
- $O: 79^{13} \bmod 2,234,807 = 2,065,927$

Encrypted HELLO: [1317460, 1261239, 1914022, 1914022, 2065927]

Step 5: Decryption ($M = C^d \bmod n$)

- $1,317,460^{171 \cdot 329} \bmod 2,234,807 = 72 = H$
- $1,261,239^{171 \cdot 329} \bmod 2,234,807 = 69 = E$
- $1,914,022^{171 \cdot 329} \bmod 2,234,807 = 76 = L$
- $1,914,022^{171 \cdot 329} \bmod 2,234,807 = 76 = L$
- $2,065,927^{171 \cdot 329} \bmod 2,234,807 = 79 = O$

Decrypted result: HELLO ✓

Question 6: Attacks on Digital Signatures [10 Marks]

1. Key-Only Attacks:

- Attacker has access to public key only
- Attempts to forge signatures without additional information

2. Known Message Attacks:

- Attacker has valid message-signature pairs
- Tries to forge signatures for new messages

3. Chosen Message Attacks:

- Attacker can obtain signatures for chosen messages
- Uses these to forge signatures for other messages

4. Hash Function Attacks:

- Collision attacks on underlying hash function
- Birthday attacks to find hash collisions

5. Implementation Attacks:

- Side-channel attacks (timing, power analysis)
- Fault injection attacks during signature generation

6. Man-in-the-Middle Attacks:

- Interception and modification of signatures
- Public key substitution attacks

7. Replay Attacks:

- Reusing valid signatures in different contexts
- Timestamp manipulation

8. Social Engineering Attacks:

- Tricking users into signing malicious documents
 - Phishing for private keys
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Question 7: Merits and Drawbacks of Digital Signatures [10 Marks]

Merits:

Security Benefits:

- **Non-repudiation:** Signer cannot deny having signed the document
- **Authentication:** Verifies the identity of the signer
- **Integrity:** Ensures document hasn't been tampered with
- **Unforgeable:** Computationally infeasible to forge valid signatures

Practical Advantages:

- **Legal validity:** Legally recognized in many jurisdictions
- **Cost-effective:** Reduces paper, printing, and storage costs
- **Speed:** Instant signing and verification
- **Global reach:** Can sign documents remotely
- **Audit trail:** Maintains record of signing process

Drawbacks:

Technical Limitations:

- **Key management complexity:** Requires secure key storage and management
- **Certificate dependency:** Relies on PKI infrastructure
- **Computational overhead:** Resource-intensive operations
- **Revocation issues:** Difficulty in handling compromised keys

Practical Challenges:

- **Technology dependence:** Requires compatible software/hardware
- **User education:** Need for technical understanding

- **Backup concerns:** Risk of losing private keys
 - **Interoperability:** Compatibility issues between different systems
 - **Long-term validity:** Questions about signature validity over time
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Question 8: Criteria of Cryptographic Hash Functions [10 Marks]

Essential Criteria:

1. Deterministic:

- Same input always produces same hash output
- Ensures consistency and reproducibility

2. Fixed Output Length:

- Hash produces fixed-size output regardless of input size
- Enables uniform storage and comparison

3. Efficient Computation:

- Fast to compute hash for any input
- Enables practical implementation

4. Pre-image Resistance (One-way property):

- Computationally infeasible to find input from hash output
- Given h , finding x where $h = \text{hash}(x)$ should be impossible

5. Second Pre-image Resistance:

- Given input x_1 , infeasible to find different x_2 where $\text{hash}(x_1) = \text{hash}(x_2)$
- Prevents targeted collision attacks

6. Collision Resistance:

- Infeasible to find any two different inputs with same hash
- Strongest security requirement

Additional Desirable Properties:

7. Avalanche Effect:

- Small input change causes significant output change

- Enhances security and unpredictability

8. Uniform Distribution:

- Hash values should be uniformly distributed
- Prevents clustering and improves security

9. Pseudorandomness:

- Output should appear random to statistical tests
- Enhances cryptographic strength

Security Implications: These criteria ensure hash functions are suitable for digital signatures, password storage, integrity verification, and other cryptographic applications. Violation of any criterion can lead to serious security vulnerabilities.